Energy Efficiency (2011) 4:321–333
DOI 10.1007/s12053-011-9108-8

Development of energy conservations scenarios for sponge iron industry using process integration

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Received: 20 March 2010 / Accepted: 3 January 2011 / Published online: 16 January 2011
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Abstract This paper presents an approach to find the best modification in the coal-based sponge iron process with an aim to integrate energy in it. For this purpose, a few energy conservation cases are formulated by integrating unutilized energy present in possible areas of process flow sheet. When unutilized energy of the process is properly integrated, energy demand from outside sources can be reduced. It decreases coal consumption as coal is the only source of energy in this plant. For energy integration in sponge iron process pinch technology is applied on the actual data of plant. Based on this data, seven energy conservation cases are identified. Among these, two are selected for detailed analysis based on utility requirement. For these two cases capital investment required for energy integration, coal consumption, water requirement, energy consumption, profit, and payback period are compared. Based on these factors the best case is selected. The integrated design for the best case includes air preheating and cooling of kiln outlet using waste gas. Further, a modified case is discussed with preheating of feed to rotary kiln and air using waste gas. This case consumes 12.3% and 93.7% less energy and water, respectively, and gives 8.6% more profit in comparison to the previous case. Thus, the modified design is selected as the best option for energy integration. This design also satisfies the practical constraints of the process.

Keywords Sponge iron plant · Energy integration · Coal consumption · Capital cost · Payback period

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Nomenclature

C	Specific heat, J/kg °C
D	Diameter of kiln, m
G	Gas
h	Heat transfer coefficient, kJ/h m^2
L	Length of kiln, m
m	Mass flow rate, kg/h
Q	Heat, kW
S	Solid
T	Temperature
t	Tonne (=1,000 kg)
NHV	Net heating value, kJ/kg
CC	Capital cost, Rupees

Subscript

A	Air
c	Coal

m	Moisture
p	Process
hu	Hot utility
s	Ore, supply
t	Target
r	Radiation
loss	Loss from kiln
i	Inlet to kiln

Greek

λ Latent heat of vaporization, kJ/kg

Introduction

Sponge iron is a metallic mass with honeycomb structure having minute holes all over the surface. Since last few years, sponge iron has emerged as an alternative raw material for steel making through the electric arc furnace (EAF) route. Seeing its tremendous potential as an alternative route to iron and steel making, the government has recognized sponge iron as vital sector for growth of Indian steel industry. Again, India is fortunate to have high reserves of iron ore of good quality and large resources of non-coking coal.

Sponge iron manufacturing unit looked very profitable since the beginning of the 1990s. However, with the liberalization of import duties of scrap, increase in input cost of raw material, and decrease in selling price, sponge iron industry found it difficult to survive in the market. Even today, when the situation has improved, sponge iron industry is undergoing several problems like lack of optimization of main equipments and energy savings schemes, etc. So, this field appears very relevant and needs further research and development.

Many investigators such as Agrawal et al. (2000, 2001), Rani Devi and Mazumder (2007), and Misra and Ipicol (2006) considered sponge iron manufacturing process and suggested improvement in it. It is found that during operation in coal-based sponge iron plant a tremendous amount of heat is generated. A significant part of this heat is associated with the waste gas and remains unutilized.

Jena et al. (1996) considered this fact and proposed a quantitative analysis based on waste gas generated, dust loss, air requirement, and efficiency of the process. They found that due to chemical reactions and combustion, the heat generated inside the kiln is 174.28 GJ/h and the heat value of coal input is 323.2 GJ/h resulting in a thermal efficiency of the process to be 53.9%. Considerable amount of heat is lost in waste gas which is about 33% of heat generated in the kiln. Authors suggested that heat in waste gas may be recovered by putting a waste heat recovery system for generation of power.

Bandyopadhyay et al. (1987) discussed that if very high volatile matter coal was used, most of the matter in feed coal was lost at the kiln mouth when coal encounters countercurrent hot waste gas. The major problem mentioned was that 30–40% of total energy goes with waste gas. Elsenheimer and Serbent (1988) also pointed out this factor. They proposed a number of options to recover energy of waste gas but did not show how much energy can be saved by installing these options and what their practical implications are.

Hajidavalloo and Alagheband (2008) studied thermal analysis of sponge iron preheating using waste energy of EAF. To improve the performance of EAF, a new initiative technique has been introduced in which sponge iron particles are preheated before entering the furnace. Based on simulation results, it is found that energy consumption in EAF can be reduced up to 14% and productivity can be increased up to 13%.

Biswas et al. (2003) and Eriksson and Larsson (2005) suggested improvements in energy efficiency by modifying rotary kiln design. Recently, Mignard and Pritchard (2007) reviewed sponge iron process for the storage and transmission of remotely generated marine energy.

The literature indicates that for Indian sponge iron industries power consumption ranges from 45 to 130 kWh/t (Ministry of Environment and Forests 2007). Significant improvements in decreasing energy consumption were achieved in gas-based processes. However, for coal-based processes potential for such savings is required. Although many plants have acquired desired level of operational efficiency but from an energy point of view various units is below optimum limit. The principal cost factor in coal-based process is energy cost as energy requirement for rotary kiln ranges between 14.63 and 20.9 GJ/t (Ulrich and Tandon 1988). It is therefore understandable that energy questions are always important in these plants.

Based on above discussion, it is observed that for heat recovery many authors considered only heat available with waste gas; however, in the sponge iron process there are many potential areas where unutilized heat is present. Thus, a fresh look is required on the existing process to integrate energy effectively. For this purpose a number of cases or modifications are identified based on the actual data of sponge iron plant. These cases are compared based on capital investment required for modification, coal consumption, water requirement, energy consumption, total profit, payback period, etc, and consequently, the best modification is selected.

Coal based sponge iron plant: a conventional process

The process flow diagram of conventional coal-based sponge iron plant is shown in Fig. 1. The operating data shown in Fig. 1 is taken from a typical Indian sponge iron plant. In the conventional process kiln feed, which consists of iron ore, feed coal, and dolomite, is fed to the rotary kiln. A separate conveyor collects different size fractions of coal for injection into the kiln with the help of pressurized air from discharge end side. Throughout length of the kiln air is injected through air fans and each of them can be adjusted separately. Further, air is injected to the kiln at the discharge end by central burner pipe, which during normal operation serves as process air inlet as shown in Fig. 2.

The inner side of a kiln is lined with refractory and supported on three piers called support rollers with an inclination of 2.5°. Rotation is given by the girth gear. Due to its inclination and rotation material in the bed of the kiln moves along the axis. Both end of the rotary kiln is provided by mechanical air sealing so that no ingress of air takes place. As charge moves through the kiln it is heated by gases, which flow in opposite direction to charge.

The first section, approximately half of the kiln, is called preheating zone wherein iron ore, coal, and dolomite are heated to reaction temperatures and the second half is the reduction zone as shown in Fig. 3. In the preheating zone, moisture of materials is driven off along with waste gas. One part of volatiles of coal escapes into the gas space above the material bed where it burns and the other part burn directly within

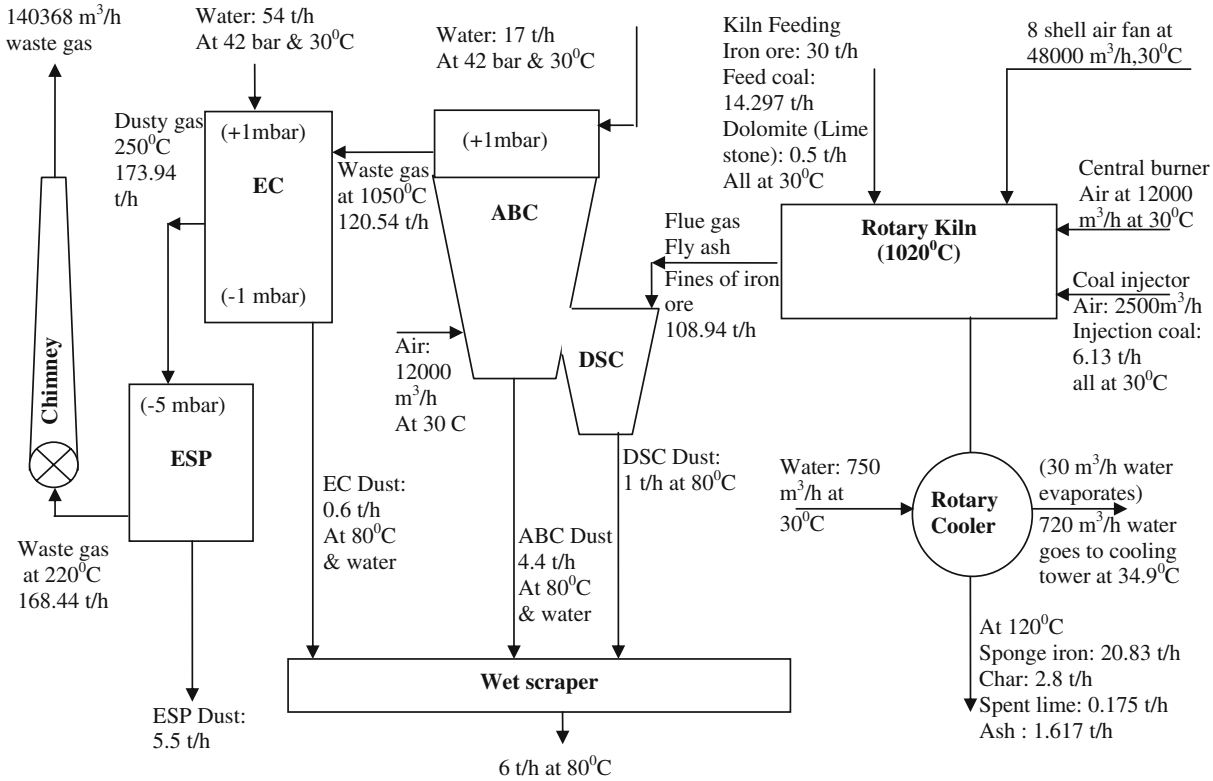


Fig. 1 Conventional sponge iron plant flow diagram

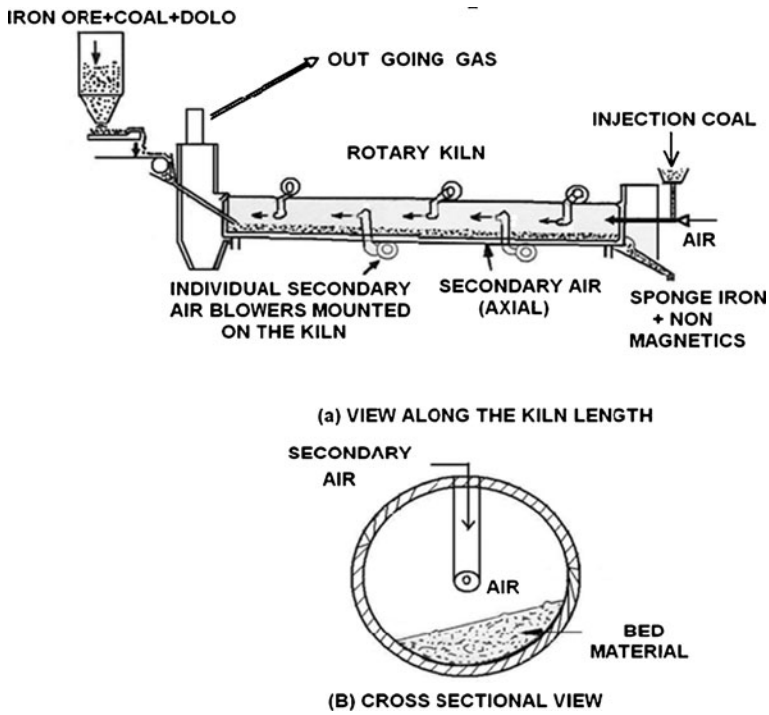


Fig. 2 Normal injection of air into the kiln

the material charge where these are used for direct heating. The volatiles in the gas space are burned by air admitted through air pipes and thus supply the main source of energy required for heating up the kiln charge. In the reduction zone, major portion of oxygen contained in iron ore is removed leaving metallic iron and a few iron oxide behind. The difference of the temperature of free gas (T(gas)) and bed material (T(charge bed)) along length of the kiln is shown in Fig. 3. This difference in temperature is responsible for heating bed charge and metallization of product. Plot of the % of metallization along length of reduction zone is shown in Fig. 3. It indicates that reduction of iron ore starts in the reduction zone and required metallization of the product is achieved at discharge end of the kiln. The operating temperature range of kiln is from 900°C to 1,020°C. The reduced product stream from the kiln is

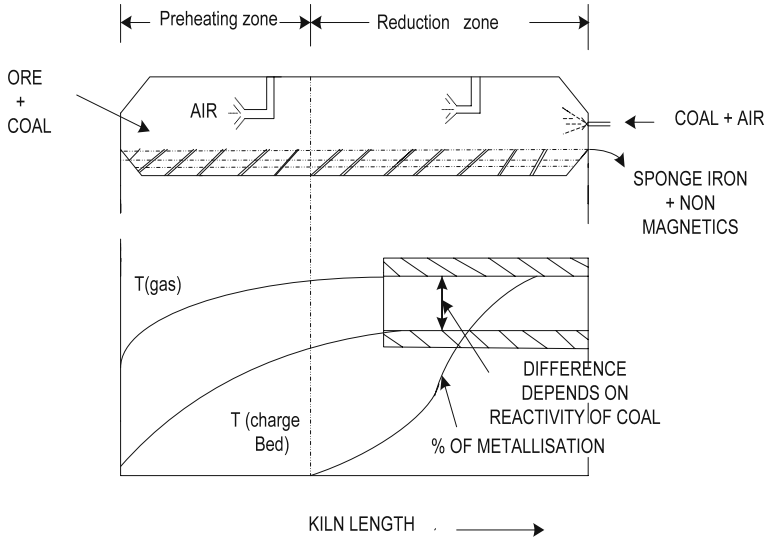


Fig. 3 Principle of counter kiln operation

indirectly cooled from 1,020°C to 120°C in a rotary cooler where water is sprayed on the cooler surface.

Waste gas in the kiln flows in opposite direction to the feed material. This flow is maintained by an induction fan mounted just before the chimney. Waste gas exiting the rotary kiln is processed through different equipment before it leaves to open atmosphere.

Waste gas consists of N_2, CO_2, CO, H_2, H_2O, O_2, and CH_4. It exits the kiln at a temperature of 900°C and then enters the after burner chamber (ABC) and a horizontal dust settling chamber (DSC) which is located beneath the ABC. DSC reduces waste gas velocity, removes large dust particles by gravity, retards pressure fluctuation, and achieves uniformity of waste gases with temperature and concentration combustible. At the end of DSC, waste gas change its direction of flow and move upward into the combustion area of ABC. Here combustibles are mixed with fresh excess air and burnt completely to acquire temperature in a range between 950°C and 1,050°C, approximately.

To remove dust particles and toxic components from waste gas, water is sprayed in the ABC. For this purpose eight to ten water guns are fitted along different heights. Water guns hold a nozzle at the discharge end for atomization. Pressurized water coming out from the guns falls on waste gas carrying dust particle and increases their weight. Subsequently, these particles settle down due to increase in weight.

At times, during the plant operation some unstable conditions are faced during which feed to the kiln is totally stopped. Under such condition, the emergency cap, which is located at the top of ABC, is operated and waste gas is allowed to escape through the emergency cap to open atmosphere.

The evaporating cooler (EC) is connected to the ABC as shown in Fig. 1. Here, waste gas is quenched as temperature of it needs to be brought down to a workable limit for downstream equipment. For quenching eight to ten water guns at same height are provided at the top position of EC around its circumference. Water, coming out from the guns, is sprayed to reduce temperature of waste gas to desired level, which is 250°C. At the bottom part of the EC, DSC and ABC is attached with a wet scraper to collect the dust.

Further, waste gas coming out from the EC enters the electrostatic precipitator (ESP) for removal of particulate matter. Desired temperature of the waste gas is to be maintained below 250°C. ESP exit is connected to the chimney through an induced direct fan. After ESP, filtered waste gas goes to the atmosphere and remaining material of ESP is collected as dust.

Cost components for 1 tonne production of sponge iron

Cost of sponge iron production varies with different inputs such as iron ore, coal, dolomite, fuel, maintenance, depreciation, power, etc. Costs of these inputs, taken directly from the plant, for 1 tonne production of sponge iron are drawn in Fig. 4, which shows that a significant component of total cost, i.e., 31.8% is due to coal.

In fact for such plants coal is the only source of energy as it is generated by combustion of coal. However, energy demand of the process can be reduced by proper integration of heat within the

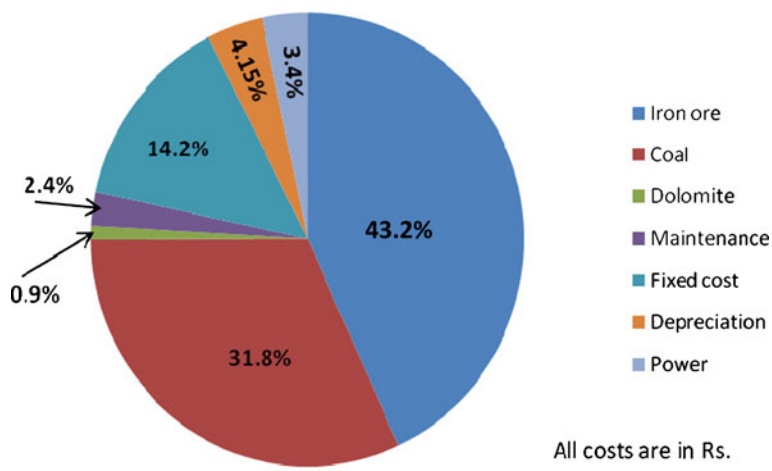


Fig. 4 Cost components for 1 tonne production of sponge iron

process which results less energy to be supplied by coal. Consequently, coal consumption is decreased which also reduces cost of coal for per tonne production of sponge iron. Thus, cost component of coal shown in Fig. 4 is directly related to the energy requirement.

Energy requirement

Sponge iron is produced by direct reduction of iron ore carried out in the rotary kiln which involves the following chemical reactions (Biswas et al. 2003):

$$1/2\ Fe_2O_3 + 3/2\ CO = Fe + 3/2\ CO_2 \quad -12,866 kJ/kmol \tag{1}$$

$$1/3\ Fe_3O_4 + 4/3\ CO = Fe + 4/3\ CO_2 \quad -3,904 kJ/kmol \tag{2}$$

$$FeO + CO = Fe + CO_2 \quad -16,108 kJ/kmol \tag{3}$$

$$FeO + C = Fe + CO \quad +156,482 kJ/kmol \tag{4}$$

Equation 4 indicates heat required for desired rate of reduction of iron oxide to iron. CO generated by Eq. 4 is utilized either for further reduction of iron ore through Eqs. 1 to 3 or for combustion ($2CO+O_2 \rightarrow 2CO_2$). It is revealed from Eq. 4 that 56 kg of sponge iron requires approximately 12 kg of carbon and 156,482 kJ/kmol of endothermic heat. Therefore, for the endothermic process heat (kJ/h) requirement is computed as:

$$Q_p = (156,482/56)\, m_s \tag{5}$$

Thus, total theoretical energy used for reduction of 30 tonnes ore is found as 6.6×10^7 kJ/kg using Eq. 5.

The actual energy consumed in a conventional plant can be computed as:

$$\text{Actual energy consumed} = m_c \times \text{NHV} \tag{6}$$

Based on Eq. 6, 4.68×10^8 kJ/kg is supplied to the process which is 7.1 times more than the theoretical value. Theoretical energy consumption is computed based on the reduction reaction involved in sponge iron process, which occurs at around 1,020°C; however, feed enters to the kiln at 30°C. In real system energy is consumed in preheating of feed (iron ore, coal, dolomite, air) up to 1,020°C, vaporization of moisture of ore and coal, radiation loss from kiln, sensible heat loss in sponge iron, char, fly ash, waste gas, volatile matter, etc. These factors cause a large difference between theoretical and actual energy consumptions. This difference advocates the basic motivation of the present study. One of the possible ways to reduce energy consumption is to integrate heat in the process and thus, minimize heat demand from outside source.

Heat integration in sponge iron plant

It is clear from above discussion that significant amount of heat is supplied to the process than the theoretical value. To reduce this gap a fresh look is required on the existing process, shown in Fig. 1 to identify possible areas where energy is being lost and can be conserved. These areas are as follows:

1. An appreciable portion of energy is being lost in the rotary kiln through untapped waste gas as has been identified by many investigators (Bandyopadhyay et al. 1987; Elsenheimer and Serbent 1988).
2. Hot sponge iron is being cooled from 1,020°C to 120°C using water in the rotary cooler. In this process vapor is generated at considerably low temperature (34.7°C) which goes directly to atmosphere.
3. Clean waste gas is generated in ESP from where it enters the chimney at a temperature of 220°C. From the chimney waste gas goes to the atmosphere.

Thus, these are a number of potential areas where energy is being lost and not tapped. This energy can be conserved and utilized in the process through proper heat integration to decrease energy bills of the industry. For this purpose, Pinch Technology (Linnhoff et al. 1982) is applied in sponge iron process. It requires necessary stream data which is extracted from the process flow sheet and presented in Table 1. Here, outlet streams of wet scrapper and ESP are considered as hot streams which cool down to ambient temperature (30°C).

Based on stream data of Table 1 a few energy conservation cases are identified. These cases are then compared for payback period, energy requirement, waste gas generation, etc. to select the best case. Possible cases based on stream data and existing system are as follows:

Table 1 Stream data of the existing process

Steam	Type	T_s	T_t	C (J/kg °C)	Flow (kg/s)	MC (kW/°C)
Kiln outlet	h1	1,020	120	678.03	7.09	4.81
Water	c1	30	34.96	4,187	208.33	872.29
Vapor	h2	34.96	33.96	1,864	8.33	15.53
Wet scrapper out	h3	80	30	881.33	1.67	1.47
ESP out	h4	125	30	896.42	1.53	1.37
Waste gas	h5	220	30	1,140	46.78	53.33

Case 1: existing system as shown in Table 1.
Case 2: existing system without vapor stream (stream h2) that is being generated in rotary cooler.
Case 3: existing system without water and vapor streams (streams c1 and h2). In this case waste gas (stream h5) behaves as cold stream and is used to cool kiln outlet (stream h1).
Case 4: existing system without dust and vapor stream (streams h2, h3, and h4).
Case 5: existing system without vapor stream (stream h2). It is different from case 2 in terms of water consumption as 375 m^3/h water (stream c1) is used in rotary cooler which is heated up from 30°C to 39.92°C.
Case 6: existing system without vapor stream (stream h2) and with air. Here, air is included in Table 1 as a cold stream from 30°C to 170°C. Kiln outlet (stream h1) is cooled using water (stream c1) at the rate 375 m^3/h in the similar manner as for case 5.
Case 7: existing system without water and vapor streams (stream c1 and h2). However, air is included in the stream data to be preheated from 30°C to 170°C using waste gas (stream h5) similar to case 6. Here cold waste gas (stream h5) is further used to cool kiln outlet (stream h1).

Utility requirements for these seven cases are computed using pinch analysis (Linnhoff et al. 1982) considering ΔT_{min} as 50°C and results obtained are tabulated in Table 2. In these cases waste gas is used either as hot stream or cold stream. Due to lower heat transfer coefficient of waste gas stream it will require substantially large surface area for heat transfer. However, the effect of lower heat transfer area can be somewhat compensated by increasing temperature difference (ΔT) between waste gas and other streams. Thus, large ΔT is preferred for operation (Linnhoff et al. 1982) when gas streams are involved in heat exchange. Due to above fact, ΔT_{min} as 50°C is considered for heat integration in all cases.

In sponge iron process variation in temperature of iron ore, coal, dolomite, sponge iron, and waste gas is in the range from 1,020°C to 30°C. Though specific heats of these materials vary considerably with change in temperature, in the present work these variations are taken as negligible. The reason behind this assumption is that the present work deals with the comparative study among different cases in terms of energy consumption where variation of specific heat due to change in temperature cancel out as temperature range in all cases is almost same.

As shown in Table 2, these cases offer threshold problem as no hot utility is required. In these cases cold utility is not used for heat integration as streams which require cold utility releases its heat directly to the atmosphere. Thus, in the present paper a modified pinch technology is used where the only hot utility is considered. This fact is also taken in to account while developing the model for computation of coal consumption as shown in "Model of coal consumption for process".

Based on utility requirements, shown in Table 2, a few cases can be selected for further analysis. As hot

Table 2 Utility requirements for seven cases

Cases	Hot utility, kW	Cold utility, kW
Case 1	0	9,869.2
Case 2	0	9,854.1
Case 3	0	198.8
Case 4	0	9,782.3
Case 5	0	9,858.4
Case 6	0	199.8
Case 7	0	198.8

utility is zero, the minimum requirement of coal utility can be a selection criterion as fewer amount of cold utility releases less heat to atmosphere. Thus, based on minimum cold utility requirement, cases 3 and 7 are considered for further analysis. It should be noted that for heat integration purpose, streams h3 and h4 are excluded from the design as these streams are dust streams and may foul the surface of heat exchanger considerably and make it inoperable in a short span of time.

Model of coal consumption for process

In the sponge iron plant, the only source of energy is coal which is produced by its combustion in the presence of air. So, coal consumption is decided by energy demand of the process which depends on hot utility requirement, heat gained by the incoming air and bed of materials, heat required for the endothermic reactions process for reduction to continue, heat lost through the rotary kiln wall, heat gained by coal, and latent heat required by moisture of feed material to evaporate. Estimation of total heat requirement for sponge iron production process is shown through following expressions:

Hot utility requirement

In the present work hot utility requirement, Q_{hu}, is computed using pinch analysis (Linnhoff et al. 1982). This utility is supplied by the combustion of coal.

Heat gained by inlet air and bed of material

In the preheating zone inlet air and iron ore is heated up to 1,020°C and then reduction takes place in respective zone. Sensible heat gained by air and ore is supplied by combustion of coal and computed using the following equations:

$$Q_a = m_a C_a \left(T_p - T_i\right)$$

$$Q_s = m_s C_s \left(T_p - T_i\right)$$

Heat required for the reduction process to continue

Heat required for reduction process is computed using Eq. 5.

Heat lost through the kiln wall and refractory

$$Q_{loss} = 2\pi DL \times h_r$$

Length and diameter of the kiln are 80 and 5 m, respectively. Actual heat lost through the wall includes heat lost from the kiln shell, inlet and outlet hoods, post combustion chamber, and inlet area of the cooler. Therefore, total heat loss is assumed as twice as that of the kiln (Jena et al. 1996).

Heat gained by coal

Sensible heat required for preheating of coal is:

$$Q_c = m_c C_c \left(T_p - T_i\right)$$

Heat required for removing moisture of feed material

The average moisture contents in coal and iron ore are 13% and 2%, respectively, by weight in Indian conditions (Agrawal et al. 2000) and heat required to remove it is computed as:

$$Q_m = m_m \times \lambda$$

$$Q_m = (0.02\, m_s + 0.13 m_c) \times \lambda$$

Combustion efficiency is assumed as 70%. It is considered as 70% of total fixed carbon available in non-coking coal that burns completely to give out heat and the remaining 30% does not burn. The major fraction of this unburnt carbon is lost to the atmosphere through waste gas as smoke and rest is discharged with sponge iron from the rotary kiln. The final empirical relation for estimating coal requirement is:

$$Q_{hu} + Q_s + Q_a + Q_p + Q_{loss} + Q_c + Q_m = m_c \mathrm{NHV}\,(0.7)$$

Or

$$Q_{hu} + Q_s + Q_a + Q_p + Q_{loss} + 0.02\, m_s 1 = m_c \left[\mathrm{NHV}\,(0.7) - C_c\left(T_p - T_i\right) - 0.13 \times \lambda\right] \quad (7)$$

The value of m_c can be predicted from Eq. 7 considering NHV of coal as 22,930.5 kJ/kg. The

property data of air, ore, and coal are constant and taken from literature (Green and Perry 2008; Grønvold and Sawelsen 1975).

Results and discussion

Coal consumption for cases 3 and 7 are computed using Eq. 7 and results are summarized in Table 3. For these cases, air of same amount, as shown in Fig. 1, is considered. In fact, in the process air requirement for combustion depends on coal consumption so when it reduces for a process inlet air requirement also decreases. Based on trial and error method actual amount of air is computed which gives actual coal consumption as shown in Table 3.

In this process operating cost depends on coal and water requirements. The costs of coal and water are Rs 2,500/t and Rs 60/t, respectively. Based on these values operating cost for all cases are computed and presented in Table 3. It shows that maximum saving of 5.4%, 91.3%, and 47.6% are found in coal consumption, water requirement, and operating cost for case 7 in comparison to the existing system.

While integrating heat in cases 3 and 7, few additional equipments are required which are shown in Table 4. Capital costs of S-G heat exchanger (double pipe) and G-G exchanger (shell and tube), shown in Table 4, are taken from the work of Prasad et al. (2010, communicated) and Shenoy (1995), respectively. In all these cases production of sponge iron remains constant, thus, reduced amount of coal as well as water give total gain in the process. Profit as well as payback period of two cases are also shown in Table 4.

In Table 4 payback periods of two cases are reported without considering the installation cost of exchangers as the present paper deals with the comparative study of different cases and results of comparison is not affected by installation cost. This fact can be explained as: cases 3 and 7 requires one and two heat exchangers of 4,480.2 and 7,658.3 kW. Based on number of heat exchangers as well as total load, it can be thought that installation costs of these exchangers are in the following order: case 3 < case 7. In Table 4 a similar trend is observed for payback period of two cases without installation cost. Thus, by considering the installation cost payback periods will be of higher value but order will remain unaltered, i.e., case 3 < case 7. So, results of comparison are not affected by considering installation cost. The results of cases 3 and 7 based on data shown in Tables 3 and 4 are discussed in the subsequent sections.

Table 3 Coal and water requirements per hour basis

Cases	Coal cons., t	Water, t	Actual coal cons., t	Operating cost, Rs
Existing	20.424	821	20.424	100,322
Case 3	20.424	71	20.424	55,322
Case 7	19.6	71	19.325	52,572

Case 3

The salient facts of case 3 are as follows:

1 In this case waste gas is used to cool kiln outlet. It reduces water requirement by 91.3% in comparison to existing system as shown in Table 3.
2 Outlet of the kiln is to be cooled from 1,020°C to 120°C which requires waste gas to be entered at 70°C as 50°C temperature difference is required for heat transfer between gas and solid (Linnhoff et al. 1982). However, waste gas of 133,110 m^3/h is available at 220°C which is cooled down to 70°C due to heat loss while passing it through a duct.
3 For this purpose, an indirect gas–solid heat exchanger is required which can be designed based on the work of Prasad et al. (2010, communicated).

Coal consumption for this case remains unchanged in comparison to existing system as integration of heat is not done between streams which are entering and exiting from the process. Here, waste gas and kiln exit are outlet streams of process. Thus, energy demand of process also remains unaltered.

Case 7

Followings are the salient features of case 7:

1 Air is preheated from 30°C to 170°C using waste gas as it is available at 220°C and 50°C temperature difference is required for transferring heat between air and waste gas (Linnhoff et al. 1982). For this purpose, 59,738.8 m^3/h waste gas

Table 4 Economic data for two cases

Cases	Water, t/h	Actual coal cons. t/h	Capital cost		Profit Rs/day	Payback, days
			Equipment required	Cost, Rs		
Case 3	71	20.424	S-G heat exchanger	39,475,288	972,000	41
Case 7	71	19.325	G-G heat exchanger	9,418,385	1,031,408	48
			S-G heat exchanger	39,475,288		

is used which is cooled down to 80°C. A G-G shell and tube heat exchanger is employed as additional equipment. Due to air preheating by waste gas 5.4% coal consumption is reduced.

2 Waste gas of 59,738.8 m^3/h cools to 80°C by supplying its heat to air. Further, this waste gas is mixed with 73,371.3 m^3/h (at 220°C) of gas. The mixture of gases is then used to cool kiln outlet. As kiln outlet is required to be cooled from 1,020°C to 120°C waste gas should be available at 70°C due to 50°C temperature difference. Waste gas achieves 70°C by moving through the duct. By exchanging heat with kiln outlet waste gas is heated from 70°C to 151.15°C. Here, 91.3% water consumption can be reduced. For this purpose S-G heat exchanger is employed as used in case 3.

For cases 3 and 7, payback periods of 41 and 48 days, respectively, are found. Based on this information, case 3 should be selected as best design. However, for case 7 coal and energy consumptions are minimized and profit is maximized. The differ-

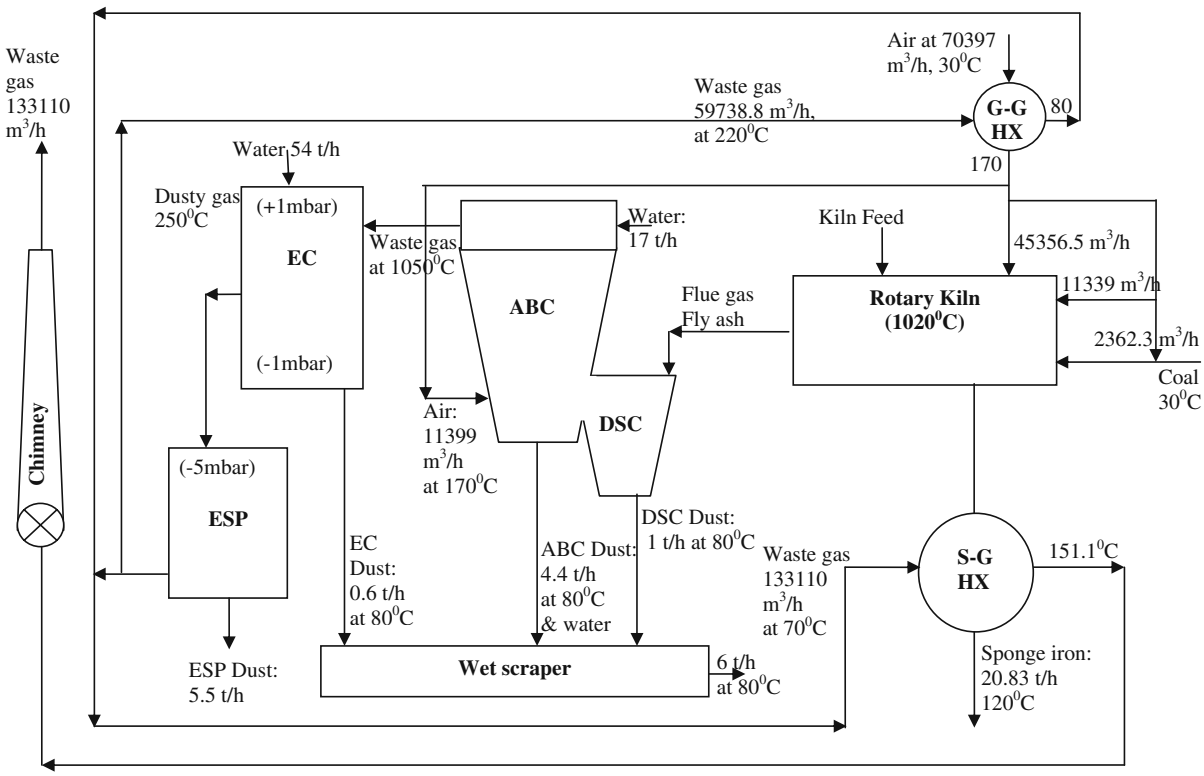


Fig. 5 Process flow diagram for case 7

Table 5 Economic analysis for all cases

Case	Coal cons.[a], t/h	Water cons., t/h	Energy cons., GJ	Operating cost, Rs/h	Capital investment		Waste gas generated, t/h	Profit, Rs/day	Payback period, days
					Item	Cost, Rs			
Existing	20.424	821	468.4	100,322	Nil	–	168.4	–	–
3	20.424	71	468.4	55,322	G-S heat exchanger	39,475,288	168.4	972,000	41
7	19.325	71	443.1	52,572	G-G heat exchanger	9,418,385	159.7	1,031,408	48
					S-G heat exchanger	39,475,288			
8	17.9	51.3	412.8	48,233	S-G heat exchanger	6,246,707	147.35	1,128,602	55
					S-G heat exchanger	1,641,248			
					S-G heat exchanger	39,475,288			
					G-G heat exchanger	14,304,704			

[a] Consumption

ence between payback periods of two cases is also not appreciable. Thus, case 7 may be selected as best option as after 7 days of operation it gives maximum profit in terms of coal and water savings. The process flow diagram for case 7 is shown in Fig. 5.

Further, it can be seen from Fig. 1 that waste gas exit the ABC is at considerably high temperature which is cooled down from 1,050°C to 250°C by spraying water in EC. However, high temperature heat of waste gas can be used in the process. For this

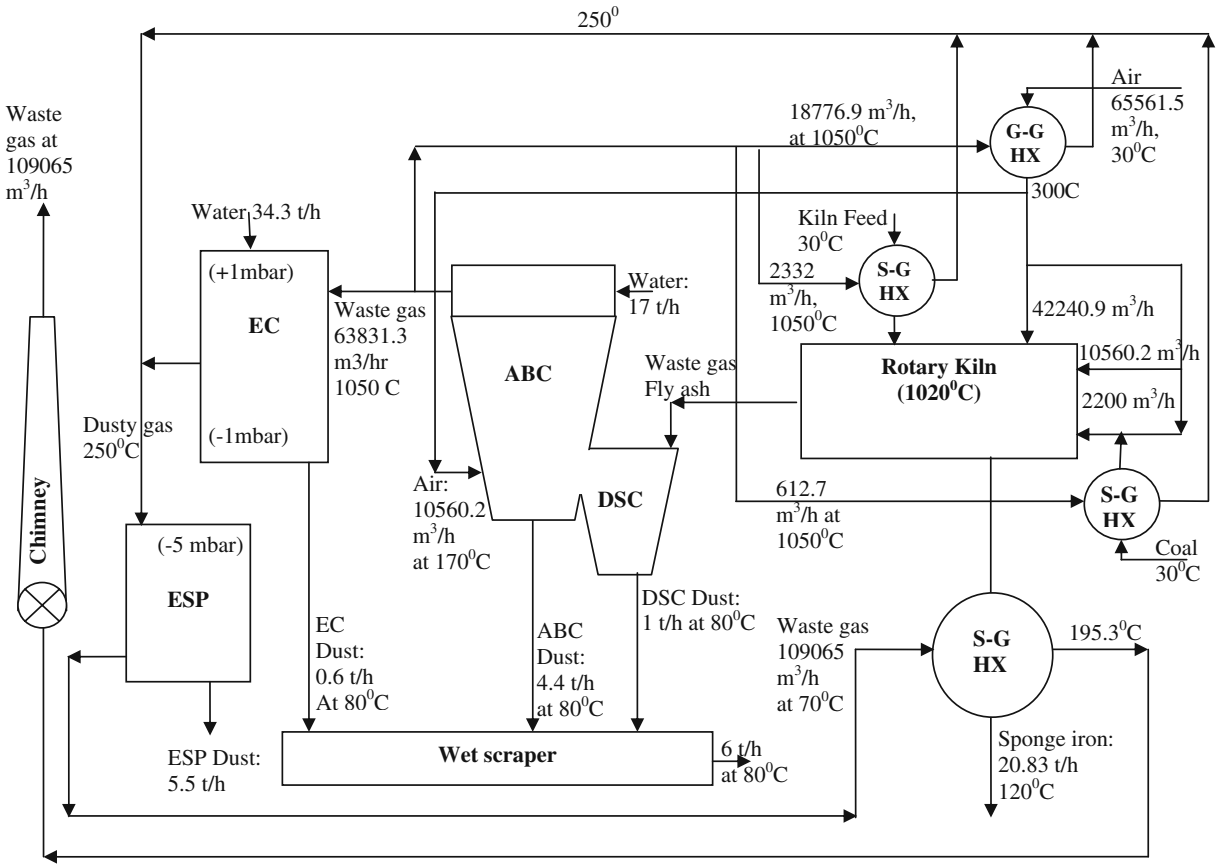


Fig. 6 Process flow diagram for case 8

purpose, a modified case is identified apart from cases 1 to 7. It is referred as case 8. For this case stream data shown in Table 1 also includes waste gas with supply temperature, target temperature, and MC value as 1,050°C, 250°C, and 53.33 kW/°C, respectively. Coal consumption for this case is computed as 17.9 t/h using Eq. 7. Features of this case are discussed in the following points:

1 Air is preheated up to 300°C using 18,776.9 m^3/h waste gas. In this case air cannot be heated more than 300°C as it may ignite coal in injector. Due to this heat exchange waste gas is cooled down from 1,050°C to 250°C. A G-G shell and tube heat exchanger is required for this purpose.
2 Further, waste gas of 2,332 m^3/h is used to preheat the feed to kiln (iron ore, feed coal, and dolomite) from 30 to 120°C which cannot be heated beyond 120°C due to conveying problem. Consequently, waste gas cools to 250°C. For this purpose S-G heat exchanger is placed as additional equipment. Design and cost of this heat exchanger is carried out in the similar manner as shown by Prasad et al. (2010, communicated).
3 Injection coal is heated from 30°C to 120°C as beyond this limit CO is generated from coal using waste gas of 612.7 m^3/h. It also requires S-G heat exchanger as additional equipment.
4 Remaining waste gas enters EC to cool down from 1,050°C to 250°C which requires 34.3 m^3/h of water.
5 Waste gas is also used to cool kiln outlet from 1,020°C to 120°C using S-G heat exchanger similar to case 3.

The modified design of case 8 consumes 93.7% and 12.3% less water and coal in comparison to the existing system. It gives total profit of Rs 1,128,602/day. However, to achieve this one G-G and three S-G heat exchangers are to be installed which gives investment of Rs 61,667,947/-. Total payback period for this case is 55 days. This modification also reduces load on EC significantly as amount of waste gas handled by it is decreased which was not the situation with case 7.

A summary of different cases with energy consumption, possible savings as well as investment is shown in Table 5. The energy requirement for cases 3, 7, and 8 are computed using Eq. 6. Table 5 shows that among all cases, case 8 consumes minimum energy, water, and coal and gives maximum profit. Though payback period is maximized for this case, it is not appreciable in comparison to case 7. Another indirect benefit of case 8 is that it releases minimum waste gas into the atmosphere. Therefore, case 8 is selected as final design for energy integrated sponge iron process plant which is shown in Fig. 6. Further, it is seen from Table 5 that for the existing system 7.1 times more energy is required than the theoretical value and by proper energy integration through case 8 this value is reduced up to 6.2.

Conclusions

The salient conclusions of the present study are as follows:

1 Based on heat integration, case 8 is selected as best design modification for sponge iron process. It consumes 93.7% and 12.3% less water and coal in comparison to existing system. It gives total profit of Rs 1,128,602/day. However, to achieve this capital investment of Rs 62,106,561/- is required. Thus, total payback period is 55 days.
2 Further, it is seen that existing system consumes 7.1 times more energy than the theoretical value. This difference is reduced up to a value of 6.2 for case 8 by proper energy integration in the process.

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